5

Morphicity and Human Competence

Situated Action Paths and Action Channels

To show how the analysis of action morphicity helps us understand how humans gain competence, we must first describe routes through action trees.[1] The recent literature on action theory suggests that an intention to act is best understood as some kind of rational commitment (Bratman 1987; Moya 1990). To form the intention today to travel to London tomorrow commits us to act in certain ways (e.g., not to buy a plane ticket to Toronto for tomorrow, not to arrange for meetings in Bath for tomorrow, and so forth). Of course such commitments can be overruled, but we would need a good reason. In addition to such "prior intentions"—that is, intentions to carry out actions at a later point in time—actors have "intentions-in-action"—that is, the intention to do what they are doing when they are doing it (Searle 1983). These intentions-in-action need not be explicitly formulated, but as we remarked in an earlier chapter, they could be stated by at least some agents if we interrupted them and asked them what they were doing.

Action trees represent sets of possible ways of carrying out a higher-level action. Faced with such an action tree, an actor must take a route through it. Carrying out a polimorphic action often involves a sequence of situated choices; after the initial commitment to the action, the agent still needs to fix, creatively, a number of parameters of the action.

1. There are some large contributions to this chapter from previously published sources. We are particularly grateful to Gerard de Vries and Wiebe Bijker for allowing us to use ideas originally published in Collins, de Vries, and Bijker (1997).

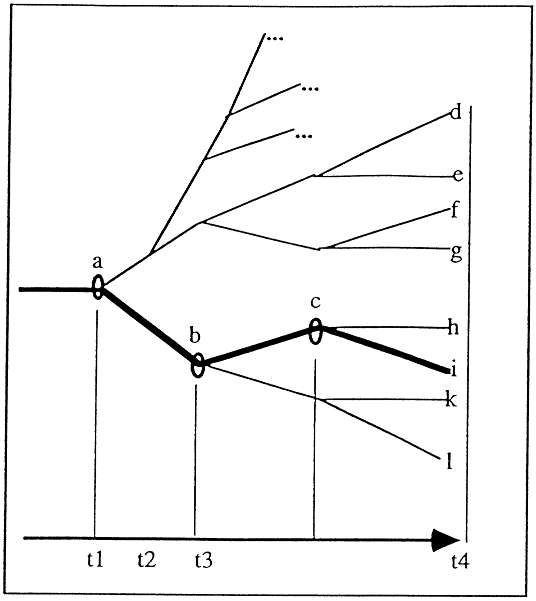

Figure 5.1
Path through an action tree.

Rotating our action trees into the conventional sideways position, we can represent this situation as follows. After the actor has formed the intention to carry out action "a," he or she still needs to make further decisions and thus needs to form further choices or "subintentions" that settle the details of the action path represented by the heavy line in figure 5.1.[2]

The term "choices" describes the route through a polimorphic action tree, but matters are subtly different in the case of mimeomorphic actions. If "a" were a *singular mimeomorphic action,* there would be no *choices* in the normal sense to be made at nodes "b" or "c"—at least, not if the action was to be executed successfully. Either there is a plan already in

2. As we explain in a later chapter, our discussion of the matter in terms of "sequence," "before," "after," and so forth can give a misleading impression of orderly temporal flow; the order is essentially logical. Sometimes all the choices will be made at once and sometimes the causal ordering of choices will run both up and down the action tree.

place that determines the route so that the whole tree could be said to collapse into a single route, or, looked at another way, and taking into account that all mimeomorphic actions have width if looked at from the appropriate frame of reference, the route through "b," "c," and so forth, is a matter of *indifference*, not choice.

If the tree represented a sequence of disjunctive mimeomorphic actions, the matter would be subtly different again. In that case the route taken at each node would not be a matter of indifference, but it would be determined in one way or another according to the circumstances pertaining at the node. To start with an easy example, if the mimeomorphic action were Taylorist chair spraying, the route would be determined by which shape of chair was passing down the production line as the actor passed across the relevant node. Thus, though the sequence of behaviors that would instantiate the actions at lower nodes may not be predictable in advance by any person, it would still not be a matter of actors' choices.

Routes through polimorphic action trees are, as we have said, determined by choices that refer to context; we will call these *situated action paths*. In contrast, routes through mimeomorphic action trees are much more like plans because they could, in principle, be specified fully without understanding the social context within which the action is set. But, since this kind of plan is not always made in advance, and since it may be beyond the capacity of humans to follow such plans even if they could be constructed, we will refer to these routes as *action channels*. Where action trees are mixed, as they usually are in real life, then the path has different characteristics at different places.

Human Competence and Machine Mimicry

Figure 5.2 shows examples of actions and their properties. We will work through each case in turn.[3] We want not simply to describe a list of competences, but to show how the idea of action morphicity helps us understand how these competences are learned, and to show that the

3. The rest of this chapter draws heavily on Collins, de Vries, and Bijker (1997), and Collins (1993).

	morph-icity 1	can be formul-ated 2	route through tree 3	how competence gained 4	level of awareness 5	skill -ful 6	can be mechan -ized 7	how mechan-ized 8
1a semaphor 1b pheasant plucker	mimeo	yes	plan	calculation	conscious	no yes	yes	program
2a marching 2b synchro-swimming	mimeo	yes	plan or channel	drill	conscious → internalised	no yes	yes	program
3 chair-spraying	mimeo	yes	channel	social skill practice	tacit	maybe	yes	research or record and playback
4 bike-balancing in the wild	mimeo	yes or potentially	channel	social skill practice	tacit	yes	yes	research or engineering
5 canoeing down rapids	mimeo	no because too complex	channel	social skill practice	tacit	yes	maybe	engineering
6 bike-riding in ordinary traffic	poli	no because socially embedded	situated action path	social skill	tacit	yes	yes	--------------
7 natural speech	poli	no because socially embedded	situated action path	social skill	tacit	no	no	--------------

Figure 5.2
Some actions and their properties.

polimorphic–mimeomorphic boundary does not correspond to other ways of dividing up our abilities.

Semaphor

In an earlier chapter we described how the word "Y-O-U" was made in semaphor. If we move along row 1a of figure 5.2, we find the properties of the action of semaphoring YOU in the various columns. Semaphoring YOU is a mimeomorphic action. The next column reminds us that a description of the action of semaphor signaling can be set down in a formula or code. The next column suggests that the route through the "semaphor YOU" action tree could be accomplished by advanced planning. Column 4 contains the claim that the kind of competence required to semaphor YOU is that required to carry out a "calculation"; for most humans there is no need to practice or drill in order to accomplish the transformation of the code into the appropriate behavior, even if they have never before encountered semaphor. Column 5 suggests that semaphoring YOU—especially if one was a beginner—would be done consciously: It could be done reasonably efficiently without "internalizing" the routine. Column 6 suggests that semaphoring YOU is *not* what we normally think of as a skillful activity. The next column is a reminder that semaphoring could be mechanized, and the next that a suitable means of mechanization would be a computer program or similar device. For example, we can imagine a mechanical semaphor signaler arranged so that the arms moved to the positions corresponding to key presses on a computer keyboard, and that this would be managed by writing a suitable program with appropriate mechanical outputs to mechanical linkages.

Case 1b: Consider the test for alcoholic intoxication that, it was said, was used by the British police before the invention of the "breathalyzer." One was asked to articulate the phrase, "The Leith police dismisseth us." Now try repeating the following at high speed and without error:

I'm not a pheasant plucker
I'm a pheasant plucker's son,
And I'm only plucking pheasant
Till the pheasant pluckers come

Both the intoxication test and the recitation of the rhyme are like semaphor in that they are mimeomorphic actions which, to be carried out successfully, require an organized, specifiable set of movements—in this case of the vocal apparatus. These tasks can be done without difficulty by existing and conceivable talking computers. The difference between these tasks and semaphor is that they are thought of as skillful performances (when carried out by drunk and sober humans respectively). Presumably this has to do with the level of self-conscious deliberation needed to accomplish them. Attempts to say "the Leith police dismisseth us" while drunk are likely to fail if uttered without thought. Likewise, said without care, the rhyme tends to have disastrous results. Here we see that mimeomorphic actions can be called skilled or unskilled and, furthermore, that there is no need for a skill to be internalized to count as a skill.

Marching

Case 2a is marching (across a parade ground). This is mimeomorphic action and its associated behaviors can be formulated (as in, say, a diagram for the position of the feet). The movements associated with marching might be planned (imagine the parade ground painted with footsteps so as to indicate the position of soldiers' feet as a stage is marked with the position of actors' feet), or it might be that one step follows from another without preplanning the whole sequence; in the later case, the route through the action tree would be an action channel because the choices at each node in the action tree would not be socially situated, they would be forced by the logic of the march. Moving to columns 4 and 5, marching is usually mastered as a result of drill; novice marchers find it hard to accomplish the action with sufficient regularity and, while initial attempts at marching would be done self-consciously, the sequence of movement associated with good marching are better done by humans when they do *not* attend to them—that is, when the sequence has been internalized. Notice, however, that we do not refer to this internalized ability as "tacit knowledge" because it is quite evident that it has come from something that could be expressed in a formula, and it could be reexpressed in a formula should we so wish. Furthermore, we do not generally say of the action of marching across a parade ground that it is a skillful activity. Moving to columns 7 and 8, should we ever want to build a marching machine there would be no special difficulty in doing it; it would involve the same sort of programming and mechanical linkages as were discussed under the topic of semaphor.

It might be possible to pick holes in many of these claims about the properties of marching, but we need to say only that there is a variant of marching (or perhaps some other activity), that fits the profile we have laid out.

Synchronized swimming (2b) contrasts with marching in the same way as the "pheasant plucker" vocalization contrasts with semaphor. In this case, *both* marching and synchronized swimming are better performed when the rules are internalized, but the swimming is an Olympic-level skill, whereas marching is not generally referred to as demonstrating high levels of skill. What we think of as skillful or not skillful does not correspond to the polimorphic–mimeomorphic boundary, nor does self-conscious versus internalized performance.

Chair Spraying

What we have in mind in case 3 is the sort of chair spraying that takes place in a closed and controlled environment (e.g., Collins 1990). Chairs are transported along an assembly line, needing to be sprayed. A human holds a spray head, directing paint onto the bare surfaces of the chair until all is covered to an appropriate depth. A limited set of different kinds of chairs may be encountered, each requiring its own set of movements.

Chair spraying is a casual mimeomorphic action, and given the possibility of different shapes of chair arriving in front of the sprayer, it is a *casual disjunctive mimeomorphic action.* There is no question that the movements of the spray head required to coat the chair could be described using some kind of formal notation, but since a human chair sprayer does not and probably could not follow such a notation in order to carry out the actions, we would have to say that the route through the chair-spraying action tree is an *action channel.* Chair spraying is learned during a kind of apprenticeship—that is, the sprayer will be shown how to do the job and will master it with a little practice, rather than do it by a calculative process. The set of movements needed to spray a chair could not be written down in such a way that they could be followed by another person who had no understanding of what the job entailed. Given this, we would probably be inclined to say that the chair sprayer had gained some tacit knowledge, though nothing turns on whether this is the correct description in this particular case. We might be inclined to call a chair sprayer a skilled person, though the task is so quickly mastered that we might not.

Notice that though another person could not master chair spraying from a set of instructions unless he understood the point of the job—which is why we say the chair sprayer has tacit knowledge—there is no question but that the job could be mechanized. It could be mechanized by analyzing the movements necessary to coat a chair, and programming a device to reproduce those movements. Or it could be done by watching a human paint sprayer very closely, redescribing his or her behaviors in terms of space–time coordinates, and proceeding with the programming as before. Or it could be done, as it is usually done, by "record and playback"; in this case the movements of the spray head are transformed, and transmitted to magnetic tape or some such, so that the whole thing can be played back without anyone having to go to the trouble of noting down the space–time coordinates.

If we accept that it is correct to describe the human paint sprayer as having tacit knowledge, then it is clear that the tacitness of knowledge is in itself no obstacle to mechanization. The next example—riding a bicycle—makes matters still more clear.

Bike Balancing

As we will see, to understand bicycle riding we need to analyze it not only in terms of a vertical action tree, but also "horizontally," because bicycle riding might mean at least two different things; these vary in terms of their meaning as actions and in terms of the extent to which they can be mechanized. In the past, when bicycle riding has been used to exemplify human skills, these aspects have been confounded.

Bicycle riding is the classic example of the use of "tacit knowledge" (Polanyi 1958). Nevertheless, so long as we stick to bike riding in a wild, traffic-free landscape (so that interactions with other humans and human conventions do not come into play), we can also agree that there is a set of mathematico-physical rules that correctly describe the dynamics of bike riding. Let us call bike riding in a wild, traffic-free landscape "bike balancing."[4]

4. This is not to say that even bike balancing makes no contact with convention. The bicycle came to look the way it does as a result of the interplay of all kinds of social forces and all kinds of disputes about what it means to ride a bike (Bijker, Hughes, and Pinch 1987). What is more, bike balancing is embedded in an action tree that, like all action trees, becomes more polimorphic the further up we go. Without the embedding in society it is almost certain that what counts as the correct way to manage a bike would not emerge. Thus a person who encountered

The set of rules for balancing a bike is either known, or potentially known, to engineers. Notwithstanding, it is clear that when we ride a bike we humans do *not* manage it by the self-conscious application of a set of mathematico-physical rules, even if we are riding across a flat surface in still conditions. We do not do it this way because we cannot judge or measure angles and speeds fast and accurately enough, nor do the sums, nor apply the results quickly enough to keep the bike upright. If we were working in .00001 of earth's gravitational field, so that things happened a hundred thousand times more slowly, or if we were a hundred thousand times quicker in our calculating than we are, we possibly could do it.[5]

Bike balancing is a relatively complicated *casual disjunctive mimeomorphic action*. It is an action that involves feedback. To repeat, the behaviors and responses that we need to carry out to keep a bike balanced are understandable and describable in physical terms, but they are still well beyond human mastery through self-conscious calculation, or

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a bicycle for the first time on a "desert island" would probably never learn to ride, for it would not occur to them that a spindly framework with two spidery wheels, surmounted by a leather blade on a stick, could be made to balance if ridden at speed. A few trials would confirm the judgment. The novice bicycle rider learns that it can be done, that it should be done, and learns how hard it is to learn, from advice and example within the embedding society (Pinch, Collins, and Carbone 1996). That there are other ways of managing a bicycle is also what makes bike balancing a formative action. That bike balancing is embedded in a polimorphic action tree does not, however, make it a polimorphic action.

Incidentally, nothing said here about the possibility of formulating a description of the physics of bike riding conflicts with ideas in the sociology of scientific knowledge. The formula may be a result of social agreements but we must still talk of a formula, for this is what a social agreement generates; there is nothing more correct than this.

5. For "Trekkies," Mr. Data is a good model. In one episode of *Star Trek* Commander Crusher shows Mr. Data the steps of a dance. She shows them only once. Mr. Data is immediately able to repeat them; Mr. Data has the kind of quick brain we are talking about. Unfortunately for *Star Trek*, Commander Crusher then tells Mr. Data to improvise, and he does this too. Improvisation is a skill requiring social embedding of a sort that Mr. Data seems to lack on other occasions. Social sensitivity is needed to know that one innovative dance step counts as an improvisation while another counts as foolishness. Thus *Star Trek* has not fully anticipated the arguments of this book.

through drill. Such skills are nearly always mastered through practice within a supporting and guiding social community.[6]

Can bike balancing be mechanized? It can, and it has. Gyrostar (1992) uses a gyroscope and attitude sensing. With something as complicated as bike balancing, mechanization is often achieved more easily through engineering solutions than through ever more ramified symbolic programming.

Canoeing Down Rapids

In one place, Suchman (1993) discusses the navigation of a canoe through a set of rapids as an example of what she calls a situated action (Suchman 1987).[7] In our terminology, we would say that canoeing down rapids is an extremely complicated *casual disjunctive mimeomorphic action.* Let us continue to be guided by the sequence of columns in figure 5.2. The action of canoeing down rapids is almost certainly too complicated to be expressed in formulaic terms, though the limitation here is one of physics; it is probably that the variations of magnitudes of forces in a fast-flowing stream are too great to foresee, and the way rocks and waves might be used to slide the canoe in different directions is equally a matter of fortuitous judgments. It is likely that the stream is impossible to model usefully, either mathematically or computationally, because of its complexity and perhaps even because there are chaotic effects rendering the links between cause and effect indeterminable, if not indeterminate. Continuing across the columns of figure 5.2, the route through the action tree of canoeing down a rapid is, in our language, an *action channel.* Canoeing is generally learned through apprenticeship with practice in a

6. As explained in previous chapters, the morphicity of actions may change over time. Bike balancing could become something else. Imagine a society in which it was thought to be bad form to ride in a state of balance all the time, and that the cute way to ride a bike was by doing "wheelies," falling off now and again, and generally "acting the clown" (see Bijker, Hughes, and Pinch 1987) for a discussion of the changing role of the bicycle). In that case, bike riding, even across a wild landscape, could not be accomplished by mimeomorphic actions because knowing the right way to ride would involve continual reference to one's audience.

7. The journal volume in which Suchman paper is found contains other examples of so-called situated actions that exhibit nothing more philosophically profound in the way of difficulty than physical complexity.

context of social support. The knowledge, once acquired, is tacit and there is no question that we refer to the ability as skillful.

Can canoeing be mechanized? It is hard to say until someone tries. No existing machine could replace the human in the canoe, but this may be just because no one has yet tried hard enough; after all, a mechanical canoeist is not something for which there is great demand. Nevertheless, from our point of view there is no principled objection to the possibility of a mechanized canoeist. Perhaps big neural nets or "situated robots" could do the job one day. Perhaps it is will be a matter of clever engineering. After all, a stick tossed in the water does quite well on its own. Not everything that cannot be mechanized cannot be mechanized because of polimorphicity; there is physical complexity too.

Bike Riding

Moving down the rows of figure 5.2, we come to a double horizontal line. Above this line all the actions are mimeomorphic, while the two below it are polimorphic. The first instance of a polimorphic action that we deal with is ordinary bicycle riding as opposed to bike balancing. Ordinary bike riding involves interaction with traffic, and traffic-controlling conventions such as traffic lights, road signs, junctions, curbs, verges, and so forth. This element of bike riding is a matter of polimorphic actions.

One might ask of a bicycle rider, "Why did you cross the junction in front of that car coming from your right?" He or she might reply: "I exchanged glances with the driver." But this does not provide a repertoire for junction crossing, since the context is so crucial. It depends on the country in which the bike is being ridden, along with an estimate of the moral integrity of its inhabitants and the particular car driver in question; it depends on whether the rider has a child on board; and, of course, it depends on the nature of the glances. Given what are otherwise the "same" circumstances, the wisdom of the move can change over time and place as the relative status of cars and bikes changes (Collins, de Vries, and Bijker 1997).

Adopting the columnar scheme once more and going to column 2, the behaviors required for ordinary bike riding cannot be formulated. Unlike the case of canoeing down rapids, however, the case is not one of the complexity of the physics, but of the social embedding of the action. The route through the action is a matter of making situated choices at nodes, so it can be neither a plan nor a channel; it is a *situated action path*. The way bike riding is mastered is undoubtedly as a social skill, and the knowledge is tacit. It is what we refer to as a skillful activity. Crucially, it cannot be mechanized unless the mechanism can be embedded somehow into the flux of social life in the same way as a bike rider is embedded. Therefore the box in the last row is blank.

Natural Speech

Ordinary language speaking is like ordinary bike riding in every respect except one and we will not bother to describe the action column by column. The one way in which speech differs from bike riding is that it is not something we normally think of as being the prerogative of the skillful person. This is in spite of its being one of the most complicated actions we ever perform.

Transference of Sets of Actions

Advice and instructions may aid the mastery of polimorphic actions, but the advice does not comprise a description of what is learned, nor can it replace experience. For a set of instructions covering a polimorphic action to be so complete that it could not be misunderstood by an unsocialized entity, it would need to anticipate all the social circumstances with which the skilled practitioner must cooperate. Neither the advice that can be comprehended by humans, nor the more complex "advice" that can be utilized by computers, amounts to a decontextualized version of a polimorphic action. Because societies cannot be simulated, all abilities below the heavy horizontal line in figure 5.2 can be transferred *only* through socialization. To learn how to interact with a society, one has to interact with that very society. We might call this the "irreplaceability thesis."

Social capabilities, like oral cultures, survive within the continued social activity of those who practice them. Social capabilities cannot be "dried out," like soup, ready to be reconstituted when immersed in the proper social context; they cannot hibernate; they can only die. That is why skills and languages disappear—barring reinvention (MacKenzie and Spinardi 1995; Pinch, Collins, and Carbone 1996).

The formula for a mimeomorphic action, on the other hand, provided it is not impossibly complex, can be inscribed in temporarily decontextualized form, and is therefore transferable in a more straightforward way. A set of mimeomorphic actions can be transferred between human contexts without being extinguished; they are "decontextualizable." We can find transferable descriptions of mimeomorphic actions in written texts, formulas, graphical plots, computer programs, and electronic or mechanical systems. The difference between polimorphic actions and mimeomorphic actions, however, is not the difference between what is "inside" humans and what is "outside." We can, as we have explained, *internalize* a set of mimeomorphic actions.

Just because the behavior associated with mimeomorphic actions can be described and stored outside of a social context, it does not follow that all mimeomorphic actions can be learned outside of social contexts. It is humans that learn to do actions, and humans have certain abilities and limitations that relate to their cognition and physiology rather than to the nature of the action itself.

There are*, then,* "*simple* mimeomorphic actions" and "complex mimeomorphic actions." Examples of repertoires of simple mimeomorphic actions include such things as making Y-O-U in semaphor or marching across a parade ground. Humans can master and even internalize repertoires of simple mimeomorphic actions away from what one might call "the scene of the action"—that is, such a repertoire of behaviors can be learned by reference to some formulaic representation, away from the location of its applications. For example, the chanting of the multiplication table—something that humans can learn as a drill—can be learned in the classroom and usefully applied outside the classroom (*pace* Lave 1988).

An example of a set of complex mimeomorphic actions is bike balancing. Even if the formula is already known to engineers, our brains are not fast enough to cope with learning to bike balance by applying the formula; normally, therefore, we learn bike balancing on a bike. Nevertheless, in contrast to the problem of simulating societies, we could build a bike balancing simulator—something like an aircraft simulator—and learn to balance a bike without ever having sat on one. It just happens that no one thinks this a worthwhile thing to do.

Simulators

The trouble with simulators is that they have to reproduce a repertoire of complex mimeomorphic actions that is the complement of what the learner has to master, and so a considerable effort of formulation is required to build them. For example, building an aircraft simulator is, in principle, something like building an autopilot. This is thought to be worthwhile only when there is a lot at stake. Given that one might drown in a canoe, one day it might just be thought worthwhile to try to build a canoeing-down-rapids simulator.

It is possible to build simulators without being able to express in a symbolic form what one is trying to simulate. Just as with mechanization, clever engineering might be an easier route. Thus to build a canoeing-down-rapids simulator, it might be easier to put a fixed canoe in a fast-flowing variable stream of water rather than computerize the whole thing as in an aircraft simulator. One can also be lucky enough to find simulators already in existence: For some aspects of surgery, animals are naturally occurring simulators for human beings.

Where simulation is not available, the only way for humans to learn sets of complex mimeomorphic actions is, as we have said, through apprenticeship in the context of use. It is the fact that repertoires of mimeomorphic actions are often learned in the same way as polimorphic actions that makes it so easy to confuse the two; the crucial point is, however, that polimorphic actions cannot be learned except through socialization or apprenticeship—mimeomorphic actions can be learned in other ways.

Conclusion

We have shown that there are three basic ways of learning to carry out an action competently. Polimorphic actions are learned through embedding within society. Simple mimeomorphic actions *can* be mastered by learning a set of body movements that can be rehearsed away from the scene of the action itself; what one might call a *drill.* More complicated disjunctive mimeomorphic actions cannot be mastered this way because they are too complex for humans to cope with; our memories are too small, our processing speed is too slow, our calculating abilities are too

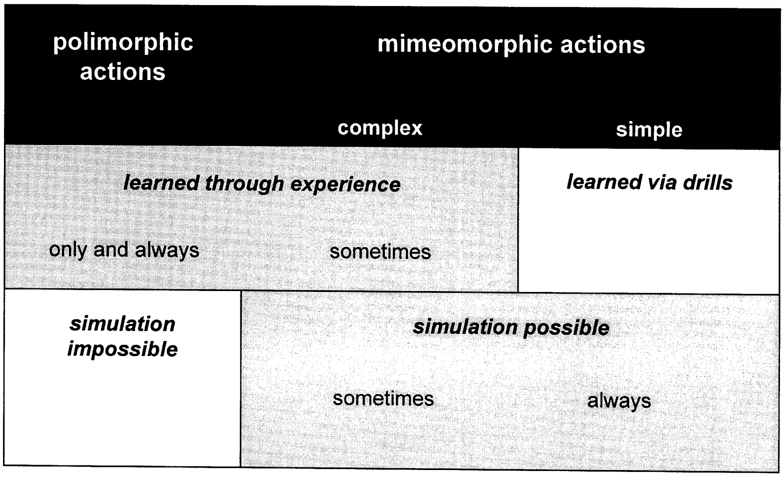

Figure 5.3
Complex and simple mimeomorphic actions

limited. *Repertoires of complex mimeomorphic actions* are usually mastered in the same way as are polimorphic actions—through practice and apprenticeship within society. Under some circumstances, however, they can be mastered through the use of simulators. The relationships are set out in figure 5.3.

Previous analyses have often worked well when the object has been to understand humans. In that case, it is sensible to lump sets of complex mimeomorphic actions along with polimorphic actions because in both cases human mastery rests upon tacit knowledge.[8] Where, however, the question is about the potential replacement without loss of humans by

8. Steven Turner's (1994) analysis of the notion of tacit knowledge suggests that it is a catchall for that which we do not yet understand. Turner may be right, regarding some uses of the term. He is, however, wrong to suggest that the idea of tacit knowledge is vacuous *because* it cannot be reduced to simple components. Turner has turned the problem on its head. Case studies have shown that knowledge maintenance and transfer cannot be reduced to matters of information (or habit) and have thus revealed that some notion like tacit knowledge—which works at the level of the collectivity—is needed if we are to understand the world. Turner's approach would dispose equally of the whole project of sociology, which rests upon the notion of culture or forms of life as the basis of the differences we see between one society and another; analyses of the consequences of these differences does not have to be held in abeyance until some reductionist program

machines, the partition needs to be put in a different place. The crucial partition is not to be located by dividing up what humans can actually do, but by considering what they might be able to do by calculation if only they were as good at processing as are computers or potential computers. If this question is asked, we find that all mimeomorphic actions, complicated or simple, lie on the opposite side of the partition from all polimorphic actions. However powerful a processor is employed, a polimorphic action cannot be mimicked by an unsocialized entity without loss. Confusion is caused, as we have pointed out, by the fact the method by which sets of complex mimeomorphic actions are mastered is often the same as that by which polimorphic actions are mastered; sets of complex mimeomorphic actions *feel like* social capabilities.

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has been fulfilled. If natural science operated in this way, it would never have started. In this chapter we decompose tacit knowledge into some of its components, without losing sight of the nonreduced aspects of that component associated with polimorphic actions.